

Diffusion processes and the Fokker-Planck equation (some ideas...)

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Linear or non linear dynamical systems

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- Discrete linear dynamical system:

$$M\ddot{q} + D\dot{q} + Kq = F(t),$$

or for $\mathbf{u} = (q, p)$, $p = M\dot{q}$:

$$\dot{\mathbf{u}}(t) = \begin{bmatrix} \mathbf{0} & M^{-1} \\ -K & -DM^{-1} \end{bmatrix} \mathbf{u}(t) + \begin{bmatrix} \mathbf{0} \\ I \end{bmatrix} F(t), \quad \mathbf{u}(0) = \mathbf{u}_0.$$

- Non-linear dynamical system, *e.g.* the Duffing oscillator:

$$M\ddot{q} + D\dot{q} + Kq + K_0q^3 = F(t),$$

or:

$$\dot{\mathbf{u}}(t) = \mathbf{b}(\mathbf{u}, t) + \mathbf{a}F(t), \quad \mathbf{u}(0) = \mathbf{u}_0,$$

$$\text{with } \mathbf{b}(\mathbf{u}, t) = \begin{pmatrix} M^{-1}p \\ -DM^{-1}p - Kq - K_0q^3 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Example: free vibrations of a single oscillator

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$$\dot{\mathbf{U}}(t) = \frac{d}{dt} \begin{pmatrix} q \\ M\dot{q} \end{pmatrix} = \underbrace{\begin{bmatrix} 0 & M^{-1} \\ -K & -DM^{-1} \end{bmatrix}}_{-\mathbf{L}} \mathbf{U}(t), \quad \mathbf{U}(0) = \mathbf{U}_0,$$

where $\mathbf{U}_0$ is a r.v. with *marginal PDF* $\pi_0(\mathbf{u}_0)$.

- Formally $\mathbf{U}(t) = \mathbf{f}(\mathbf{U}_0, t) = e^{-\mathbf{L}t} \mathbf{U}_0$ and the *marginal PDF* at time t , $\pi(\mathbf{u}; t)$, is given by the *causality principle* (lecture #1):

$$\pi_0(\mathbf{u}_0) = \pi(\mathbf{f}(\mathbf{u}_0, t)) \det(\nabla_{\mathbf{u}} \mathbf{f}).$$

- It yields the conservation equation of the PDF:

$$0 = \frac{d\pi_0}{dt} = \partial_t \pi + \nabla_{\mathbf{u}} \cdot (\pi \dot{\mathbf{u}}) = \partial_t \pi + \nabla_{\mathbf{u}} \cdot \mathbf{J}(\pi)$$

with $\mathbf{J}$ the probability flux and $\mathbf{J}(\pi) = -\pi \mathbf{L} \mathbf{u}$ the constitutive behavior.

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A general first-order stochastic differential equation for the process U indexed on $\mathbb{R}_+$ with values in $\mathbb{R}^q$:

$$\dot{U}(t) = \mathbf{b}(U, t) + \mathbf{a}(U, t)\mathbf{F}(t), \quad U(0) = U_0,$$

with the data:

- $\mathbf{u}, t \mapsto \mathbf{b}(\mathbf{u}, t) : \mathbb{R}^q \times \mathbb{R}_+ \rightarrow \mathbb{R}^q$ the *drift* function;
- $\mathbf{u}, t \mapsto \mathbf{a}(\mathbf{u}, t) : \mathbb{R}^q \times \mathbb{R}_+ \rightarrow \mathbb{M}_{q,p}(\mathbb{R})$ the *scattering* operator;
- U_0 is an r.v. in $\mathbb{R}^q$ with known marginal PDF $\pi_0(\mathbf{u}_0)$;
- $\mathbf{F}(t) = (F_1(t), \dots, F_p(t))$ is a second-order Gaussian random process indexed on $\mathbb{R}$ with values in $\mathbb{R}^p$, also centered, stationary, such that $F_1(t), \dots, F_p(t)$ are mutually independent and mean-square continuous, with:

$$\mathbf{S}_{\mathbf{F}}(\omega) = S_0 \mathbf{1}_{[-B, B]}(\omega) [\mathbf{I}_p], \quad S_0 > 0, \quad B > 0.$$

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- $B < +\infty$: colored noise, hot topic!

$$\mathbf{U}(t) = \mathbf{U}_0 + \int_0^t \mathbf{b}(\mathbf{U}(s), s) ds + \int_0^t \mathbf{a}(\mathbf{U}(s), s) \mathbf{F}(s) ds.$$

- $B \rightarrow +\infty$: $\mathbf{F} \rightarrow \dot{\mathbf{W}}$ the *normalized Gaussian white noise*, and the solution of the first-order SDE holds as a “stochastic integral”:

$$\mathbf{U}(t) = \mathbf{U}_0 + \int_0^t \mathbf{b}(\mathbf{U}(s), s) ds + \int_0^t \mathbf{a}(\mathbf{U}(s), s) \circ d\mathbf{W}(s).$$

- *Causality*: the family of r.v. $\{\mathbf{U}(s), 0 \leq s \leq t\}$ is independent of the family of r.v. $\{\mathbf{F}(\tau), \tau > t\}$ or $\{d\mathbf{W}(\tau), \tau > t\}$.

White noise

Definition

Definition

- *The normalized Gaussian white noise $\mathbf{B}(t) \equiv \dot{\mathbf{W}}(t)$ with values in $\mathbb{R}^p$ is the Gaussian stochastic process indexed on $\mathbb{R}$, centered, stationary, with the spectral density matrix:*

$$\mathbf{S}_{\mathbf{B}}(\omega) = \frac{1}{2\pi} \mathbf{I}_p.$$

- *Since $B_1(t), \dots, B_p(t)$ are uncorrelated and jointly Gaussian, they are mutually independent.*
- *$\mathbf{B}$ is not second order $\|\mathbf{B}(t)\|^2 = \int \text{Tr } \mathbf{S}_{\mathbf{B}}(\omega) d\omega = +\infty$.*

This definition holds in the sense of generalized stochastic processes $\varphi \mapsto \mathbf{B}(\varphi) : \mathcal{D}(T) \rightarrow L^2(\Omega, \mathbb{R}^p)$ where $\mathcal{D}(T)$ is the set of $\mathcal{C}^\infty$ functions having a compact support within $T \subseteq \mathbb{R}$.

White noise

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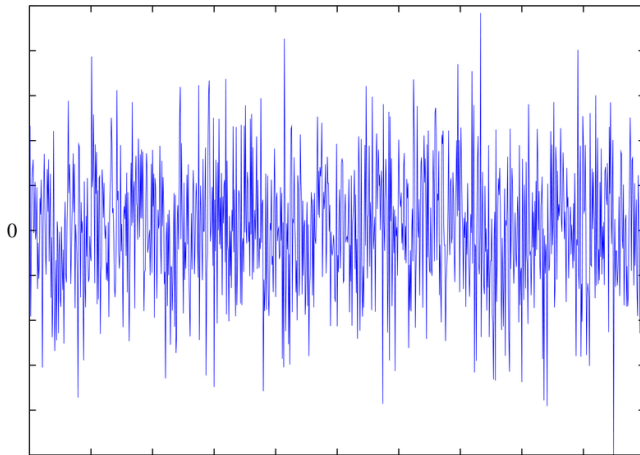
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White noise.

Wiener process

Definition

The white noise is the (generalized) derivative of the Wiener process, or *Brownian motion*.

Definition

The (normalized) Wiener process $\mathbf{W}(t)$ with values in $\mathbb{R}^p$ is the stochastic process indexed on $\mathbb{R}_+$, s.t.:

- $W_1(t), \dots, W_p(t)$ are mutually independent;
- $\mathbf{W}(0) = \mathbf{0}$ almost surely (a.s.);
- If $0 \leq s < t < +\infty$ let $\Delta \mathbf{W}(s, t) = \mathbf{W}(t) - \mathbf{W}(s)$, then:
 - $\forall m$ and $0 < t_1 < t_2 < \dots < t_m < +\infty$, $\mathbf{W}(0)$, $\Delta \mathbf{W}(0, t_1)$, $\Delta \mathbf{W}(t_1, t_2)$, ... $\Delta \mathbf{W}(t_{m-1}, t_m)$ are mutually independent r.v. (independent increments);
 - $\Delta \mathbf{W}(s, t)$ is a Gaussian, centered, second-order r.v. with $C_{\Delta \mathbf{W}}(s, t) = (t - s)\mathbf{I}_p$.

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Wiener process

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Consequently it can be shown that:

- $\mathbf{W}(t)$ is a second-order Gaussian, centered, mean-square continuous, non stationary stochastic process;
- the covariance and conditional PDF for $0 \leq t, s < +\infty$:

$$\mathbf{C}_{\mathbf{W}}(t, s) = \text{Min}(t, s) \mathbf{I}_p,$$

$$\pi_t(\mathbf{v}'; t + s | \mathbf{v}; t) = (2\pi s)^{-\frac{p}{2}} e^{-\frac{\|\mathbf{v}' - \mathbf{v}\|^2}{2s}};$$

- $\mathbf{W}(t)$ has a.s. continuous sample paths;
- sample paths $t \mapsto \mathbf{W}(t, \theta)$, $\theta \in \Omega_\theta$, are non differentiable a.s.

As a generalized derivative with $d\mathbf{W} = (dW_1, \dots, dW_p)$:

$$d\mathbf{W}(\varphi) = \mathbf{B}(-\dot{\varphi}), \quad \forall \varphi \in \mathcal{D}(\mathbb{R}).$$

Wiener process

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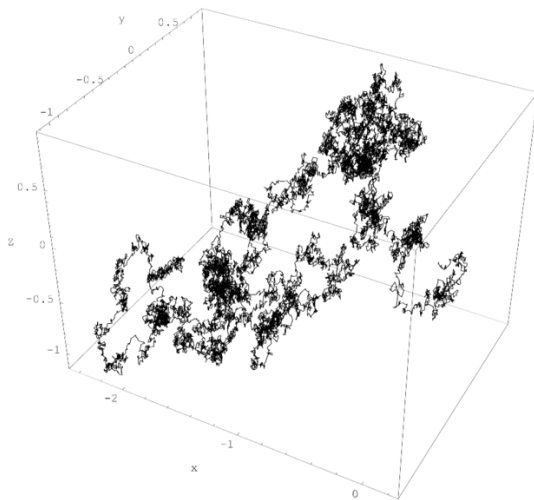
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Wiener process in $\mathbb{R}^3$.

Stochastic integrals

Definition

- Let $\mathbf{X}(t)$ be a stochastic process indexed by $\mathbb{R}_+$ with a.s. continuous sample paths.
- Assume the r.v. $\{\mathbf{X}(s), 0 \leq s \leq t\}$ are independent of the r.v. $\{\Delta \mathbf{W}(t, \tau), \tau > t\}$: a *non anticipative* process, then

$$\begin{aligned} & \int_0^t \mathbf{X}(s) d_\lambda \mathbf{W}(s) \\ &= \text{l. i. p.} \sum_{K \rightarrow +\infty}^K [(1 - \lambda) \mathbf{X}(t_k) + \lambda \mathbf{X}(t_{k+1})] \Delta \mathbf{W}(t_k, t_{k+1}), \end{aligned}$$

for any partition $0 = t_1 < t_2 < \dots < t_{K+1} = t$ of $[0, t]$ with $\max_{1 \leq k \leq K} (t_{k+1} - t_k) \xrightarrow{K \rightarrow +\infty} 0$.

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Stochastic integrals

Application to stochastic differential calculus

- A simple example—remind $\Delta W \propto \Delta t^{\frac{1}{2}}$ for the real-valued Wiener process W :

$$\int_0^t W(s) d_\lambda W(s) = \frac{1}{2} W(t)^2 + \left(\lambda - \frac{1}{2} \right) t,$$

from which one deduces the stochastic differential:

$$d_\lambda(W(t)^2) = 2W(t)d_\lambda W(t) + (1 - 2\lambda)dt.$$

- More generally ($\lambda = 0$ is called the *Itô formula*):

$$d_\lambda(f(W(t))) = f'(W(t))d_\lambda W(t) + \left(\frac{1}{2} - \lambda \right) f''(W(t))dt.$$

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- If $\lambda = \frac{1}{2}$ the usual differential calculus applies, and the solution of SDE holds as a *Stratonovich integral* (1966):

$$U(t) = U_0 + \int_0^t \mathbf{b}(U(s), s) ds + \int_0^t \mathbf{a}(U(s), s) \circ d\mathbf{W}(s).$$

- If $\lambda = 0$, its solution holds as an *Itô integral* (1944):

$$U(t) = U_0 + \int_0^t \underline{\mathbf{b}}(U(s), s) ds + \int_0^t \mathbf{a}(U(s), s) d\mathbf{W}(s),$$

where:

$$\underline{\mathbf{b}}(\mathbf{u}, t) = \mathbf{b}(\mathbf{u}, t) + \frac{1}{2} \mathbf{a}^\top \nabla_{\mathbf{u}} \mathbf{a}.$$

- $U(t)$ is a *Markov process*.

Stochastic integrals

Itô's formula

- Let $\mathbf{U}(t) \in \mathbb{R}^q$ be the solution of the ISDE:

$$\mathbf{U}(t) = \mathbf{U}_0 + \int_0^t \underline{\mathbf{b}}(\mathbf{U}(s), s) ds + \int_0^t \mathbf{a}(\mathbf{U}(s), s) d\mathbf{W}(s).$$

- Let $\phi : \mathbb{R}^q \times \mathbb{R} \rightarrow \mathbb{R}$ be a smooth function. Then *Itô's formula* states that:

$$\begin{aligned} \phi(\mathbf{U}(t), t) &= \phi(\mathbf{U}_0, 0) + \int_0^t \frac{\partial \phi}{\partial t}(\mathbf{U}(s), s) ds \\ &\quad + \int_0^t \nabla_{\mathbf{u}} \phi(\mathbf{U}(s), s) \cdot d\mathbf{U}(s) \\ &\quad + \frac{1}{2} \int_0^t \nabla_{\mathbf{u}} \otimes \nabla_{\mathbf{u}} \phi(\mathbf{U}(s), s) : \mathbf{a}(\mathbf{U}(s), s) \mathbf{a}(\mathbf{U}(s), s)^\top ds, \end{aligned}$$

where $d\mathbf{U}(t) = \underline{\mathbf{b}}(\mathbf{U}(s), s) ds + \mathbf{a}(\mathbf{U}(s), s) d\mathbf{W}(s)$.

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Markov processes

Definition

Definition

The conditional probability given $t_0 < \dots < t_m < t$:

$$\pi_t(\mathbf{u}; t | \mathbf{u}_0, \dots, \mathbf{u}_m; t_0, \dots, t_m) = \frac{\pi(\mathbf{u}_0, \dots, \mathbf{u}_m, \mathbf{u}; t_0, \dots, t_m, t)}{\pi(\mathbf{u}_0, \dots, \mathbf{u}_m; t_0, \dots, t_m)}.$$

Definition

Let $\mathbf{U}(t)$ be a stochastic process defined on $(\Omega, \mathcal{E}, P)$ and indexed on $\mathbb{R}_+$ with values in $\mathbb{R}^q$. It is a Markov process if:

- *for all $0 \leq t_1 < \dots < t_m < t$ and $\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{u}$ in $\mathbb{R}^q$*

$$\pi_t(\mathbf{u}; t | \mathbf{u}_0, \dots, \mathbf{u}_m; t_0, \dots, t_m) = \pi_t(\mathbf{u}; t | \mathbf{u}_m; t_m);$$

- *the marginal PDF $\pi_0(\mathbf{u}_0)$ of $\mathbf{U}(0)$ can be any PDF.*

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Markov processes

Chapman-Kolmogorov equation

- A Markov process is fully characterized by:
 - its marginal PDF $\pi(\mathbf{u}; t)$,
 - and its *transition PDF* $\pi_t(\mathbf{u}; t | \mathbf{v}; s)$, $0 \leq s < t < +\infty$, with

$$\pi(\mathbf{u}; t) = \int_{\mathbb{R}^q} \pi_t(\mathbf{u}; t | \mathbf{v}; s) \pi(\mathbf{v}; s) d\mathbf{v}.$$

- π_t satisfies the *Chapman-Kolmogorov equation*:

$$\pi_t(\mathbf{u}; t | \mathbf{u}'; t') = \int_{\mathbb{R}^q} \pi_t(\mathbf{u}; t | \mathbf{v}; s) \pi_t(\mathbf{v}; s | \mathbf{u}'; t') d\mathbf{v}, \quad t' < s < t.$$

- Homogeneous Markov process:

$$\pi_t(\mathbf{u}; t | \mathbf{v}; s) = \pi_t(\mathbf{u}; t - s | \mathbf{v}; 0), \quad 0 \leq s < t < +\infty.$$

- The Brownian motion is a Markov process.

Diffusion processes

Definition

Definition

The $\mathbb{R}^q$ -valued Markov process $\mathbf{U}(t)$ with a.s. continuous sample paths and transition PDF $\pi_t(\mathbf{v}; s|\mathbf{u}; t)$ is a diffusion process if $\forall \epsilon > 0$ (but not necessarily small), $\forall \mathbf{u} \in \mathbb{R}^q$ the first moments of its increments are such that for $h > 0$:

$$\int_{\|\mathbf{v}-\mathbf{u}\| \geq \epsilon} \pi_t(d\mathbf{v}; t+h|\mathbf{u}; t) = o(h),$$

$$\int_{\|\mathbf{v}-\mathbf{u}\| < \epsilon} (\mathbf{v} - \mathbf{u}) \pi_t(d\mathbf{v}; t+h|\mathbf{u}; t) = h \underline{\mathbf{b}}(\mathbf{u}, t) + o(h),$$

$$\int_{\|\mathbf{v}-\mathbf{u}\| < \epsilon} (\mathbf{v} - \mathbf{u}) \otimes (\mathbf{v} - \mathbf{u}) \pi_t(d\mathbf{v}; t+h|\mathbf{u}; t) = h \boldsymbol{\sigma}(\mathbf{u}, t) + o(h),$$

where $\underline{\mathbf{b}} \in \mathbb{R}^q$ and $\boldsymbol{\sigma} \in \mathbb{M}_{q,q}(\mathbb{R})$ symmetric, positive.

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- *Continuity*: particles moving on sample paths of a diffusion process only make small jumps, or the probability of moving a distance ϵ goes to zero as h goes to zero no matter how small ϵ is.
- *Drift*: those particles can have a net mean velocity $\underline{b}$.
- *Diffusion*: particles spread as time increases with the rate $\text{Tr } \sigma$. Entropy increases while the phase space contracts, thus some information (energy) gets lost.

$$U(t+h) - U(t) \approx h \underline{b}(U(t), t) + \sigma^{\frac{1}{2}}(U(t), t) \Delta W(t, t+h).$$

Fokker-Planck equation

The marginal PDF π and transition PDF π_t of a diffusion process satisfy the *Fokker-Planck equation*:

$$\partial_t \pi + \nabla_{\mathbf{u}} \cdot \left(\pi \underline{\mathbf{b}} - \frac{1}{2} \nabla_{\mathbf{u}} \cdot (\pi \boldsymbol{\sigma}) \right) = 0,$$

with $\pi(\mathbf{u}_0; 0) = \pi_0(\mathbf{u}_0)$ and $\lim_{h \downarrow 0} \pi_t(\mathbf{u}; t+h|\mathbf{v}; t) = \delta(\mathbf{u} - \mathbf{v})$.

$$\begin{aligned} \int_{\mathbb{R}^q} f(\mathbf{u}) \partial_t \pi_t(\mathbf{u}; t|\mathbf{v}; s) d\mathbf{u} &= \lim_{h \downarrow 0} \frac{1}{h} \int_{\mathbb{R}^q} f(\mathbf{u}) (\pi_t(\mathbf{u}; t+h|\mathbf{v}; s) - \pi_t(\mathbf{u}; t|\mathbf{v}; s)) d\mathbf{u} \\ &= \lim_{h \downarrow 0} \frac{1}{h} \int_{\mathbb{R}^q} \pi_t(\mathbf{u}; t|\mathbf{v}; s) \left[\int_{\mathbb{R}^q} f(\mathbf{u}') \pi_t(\mathbf{u}'; t+h|\mathbf{u}; t) d\mathbf{u}' - f(\mathbf{u}) \right] d\mathbf{u} \quad (\text{C-K}) \\ &= \lim_{h \downarrow 0} \frac{1}{h} \int_{\mathbb{R}^q} \pi_t(\mathbf{u}; t|\mathbf{v}; s) \int_{\mathbb{R}^q} (f(\mathbf{u}') - f(\mathbf{u})) \pi_t(\mathbf{u}'; t+h|\mathbf{u}; t) d\mathbf{u}' d\mathbf{u} \quad (\text{norm.}) \\ &= \lim_{h \downarrow 0} \frac{1}{h} \int_{\mathbb{R}^q} \pi_t(\mathbf{u}; t|\mathbf{v}; s) \int_{\|\mathbf{u}' - \mathbf{u}\| < \epsilon} (f(\mathbf{u}') - f(\mathbf{u})) \pi_t(\mathbf{u}'; t+h|\mathbf{u}; t) d\mathbf{u}' d\mathbf{u}, \quad \forall f \in \mathcal{C}_0^2. \end{aligned}$$

Then use a Taylor expansion for f , definitions of drift and diffusion, and integrate by parts.

Itô's stochastic differential equations (ISDE)

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$$dU = \underline{b}(U, t)dt + \underline{a}(U, t)dW, \quad U(0) = U_0,$$

with the regularity assumptions:

$$\begin{aligned} \|\underline{b}(\mathbf{u}, t)\| + \|\underline{a}(\mathbf{u}, t)\| &\leq K(1 + \|\mathbf{u}\|), \\ \|\underline{b}(\mathbf{u}', t) - \underline{b}(\mathbf{u}, t)\| + \|\underline{a}(\mathbf{u}', t) - \underline{a}(\mathbf{u}, t)\| &\leq K\|\mathbf{u}' - \mathbf{u}\|. \end{aligned}$$

- 1 Then the SDE has a unique solution, with a.s. continuous sample paths. If in addition $\underline{b}$ and $\underline{a}$ are independent of t , $U(t)$ is homogeneous.
- 2 If $t \mapsto \underline{b}(\mathbf{u}, t)$ and $t \mapsto \underline{a}(\mathbf{u}, t)$ are continuous, $U(t)$ is also a diffusion process with $\sigma = \underline{a}\underline{a}^\top$.

Itô's stochastic differential equations (ISDE)

Example: Black-Scholes¹ model

- The relative variation of a stock $U(t)$ with constant (annualized) drift rate μ and volatility σ :

$$\frac{dU}{U} = \mu dt + \sigma dW, \quad U(0) = U_0.$$

- Transformation to a Stratonovich SDE:

$$\frac{dU}{U} = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma \circ dW, \quad U(0) = U_0,$$

for which “normal rules of integration” apply:

$$U(t) = U_0 e^{\sigma W(t) + (\mu - \frac{\sigma^2}{2})t}.$$

- The Fokker-Planck equation:

$$\partial_t \pi + \mu \partial_u (\pi u) - \frac{\sigma^2}{2} \partial_u^2 (\pi u^2) = 0, \quad \pi(u; 0) = \pi_0(u).$$

¹Fischer Black (1938–1995), Myron Scholes (1941–): American financial economists. M. Scholes received the Sveriges Riksbank Prize in Economic Sciences in Memory of A. Nobel in 1997 for this model for valuing options, together with Robert Merton (1944–).

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A general first-order stochastic differential equation for the process U indexed on $\mathbb{R}_+$ with values in $\mathbb{R}^q$:

$$\begin{cases} \dot{U}(t) = \mathbf{b}(U, t) + \mathbf{a}(U, t)\mathbf{F}(t), & t > 0, \\ U(0) = U_0, \end{cases}$$

with the data:

- $\mathbf{u}, t \mapsto \mathbf{b}(\mathbf{u}, t) : \mathbb{R}^q \times \mathbb{R}_+ \rightarrow \mathbb{R}^q$ the *drift* function;
- $\mathbf{u}, t \mapsto \mathbf{a}(\mathbf{u}, t) : \mathbb{R}^q \times \mathbb{R}_+ \rightarrow \mathbb{M}_{q,p}(\mathbb{R})$ the *scattering* operator;
- U_0 is an r.v. in $\mathbb{R}^q$ with known marginal PDF $\pi_0(\mathbf{u}_0)$;
- $\mathbf{F}(t) = (F_1(t), \dots, F_p(t))$ is a second-order Gaussian random process indexed on $\mathbb{R}^+$ with values in $\mathbb{R}^p$, centered, mean-square continuous.

Markovian realization

Definition

Definition

$\mathbf{F}(t)$ indexed on $\mathbb{R}^+$ with values in $\mathbb{R}^p$, second-order, Gaussian, centered and mean-square continuous admits a Markovian realization if:

$$\left\{ \begin{array}{ll} \mathbf{F}(t) = \mathbf{H}\mathbf{V}(t), & t \geq 0, \\ \dot{\mathbf{V}}(t) = \mathbf{P}\mathbf{V}(t) + \mathbf{Q}\mathbf{B}(t), & t > 0, \\ \mathbf{V}(0) = \mathbf{V}_0 & a.s. \end{array} \right.$$

where $\mathbf{V}_0$ is a Gaussian r.v. in $\mathbb{R}^n$, $\mathbf{V}(t)$ is a diffusion process indexed on $\mathbb{R}_+$ with values in $\mathbb{R}^n$, $\mathbf{P}, \mathbf{Q} \in \mathbb{M}_n(\mathbb{R})$, $\mathbf{H} \in \mathbb{M}_{p,n}(\mathbb{R})$, $\Re\{\lambda_j(\mathbf{P})\} < 0$.

- This is equivalent to a linear Itô stochastic differential equation.
- $\mathbf{V}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}_0)$ where $\mathbf{\Sigma}_0 = \int_0^{+\infty} e^{\tau\mathbf{P}} \mathbf{Q}\mathbf{Q}^\top e^{\tau\mathbf{P}^\top} d\tau$.

Physically realizable process

Definition

Definition

$\mathbf{F}(t)$ indexed on $\mathbb{R}$ with values in $\mathbb{R}^p$, second-order, mean-square stationary and continuous, centered, is physically realizable if $\exists \mathbb{H} \in L^2(\mathbb{R})$, $\text{supp } \mathbb{H} \subseteq \mathbb{R}_+$, s.t.:

$$\mathbf{F}(t) = \int_{-\infty}^t \mathbb{H}(t - \tau) \mathbf{B}(\tau) d\tau,$$

or equivalently $\mathbf{S}_{\mathbf{F}}(\omega) = \frac{1}{2\pi} \hat{\mathbb{H}}(\omega) \hat{\mathbb{H}}(\omega)^, \forall \omega \in \mathbb{R}$.*

A necessary and sufficient condition (Rozanov 1967):

$$\int_{\mathbb{R}} \frac{\ln(\det \mathbf{S}_{\mathbf{F}}(\omega))}{1 + \omega^2} d\omega > -\infty.$$

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Markovian realization

Existence for a physically realizable process

Theorem

A necessary and sufficient condition:

$$\mathbf{S}_{\mathbf{F}}(\omega) = \frac{\mathbf{R}(\mathrm{i}\omega)\mathbf{R}(\mathrm{i}\omega)^*}{2\pi|P(\mathrm{i}\omega)|^2}, \quad \text{or} \quad \mathbb{H}(\omega) = \frac{\mathbf{R}(\mathrm{i}\omega)}{P(\mathrm{i}\omega)},$$

where:

- $P(z)$ is a polynomial of degree d on $\mathbb{C}$ with real coefficients and roots in the half-plane $\Re(z) < 0$,
- $\mathbf{R}(z)$ is a polynomial on $\mathbb{C}$ with coefficients in $\mathbb{M}_{p,n}(\mathbb{R})$ and degree $r < n$.

The Markovian realization always exists in infinite dimension $n = +\infty$.

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A non linear first-order stochastic differential equation for the process $\mathbf{Z}(t) = (\mathbf{U}(t), \mathbf{V}(t))$ indexed on $\mathbb{R}_+$ with values in $\mathbb{R}^\nu$, $\nu = q + n$:

$$\begin{cases} d\mathbf{Z}(t) = \mathbf{b}_z(\mathbf{Z}, t)dt + \mathbf{a}_z d\mathbf{W}, & t > 0, \\ \mathbf{Z}(0) = \mathbf{Z}_0, \end{cases}$$

where $\mathbf{Z}_0 = (\mathbf{U}_0, \mathbf{V}_0)$,

$$\mathbf{b}_z(\mathbf{u}, \mathbf{v}, t) = \begin{bmatrix} \mathbf{b}(\mathbf{u}, t) + \mathbf{a}(\mathbf{u}, t)\mathbf{H}\mathbf{v} \\ \mathbf{P}\mathbf{v} \end{bmatrix}, \quad \mathbf{a}_z = \begin{bmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} \end{bmatrix},$$

and $\mathbf{W}(t)$ is the Wiener process in $\mathbb{R}^\nu$.

Numerical integration of SDE

Strong convergence

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$$\begin{cases} dU(t) = b(U, t)dt + a(U, t)dW(t), & t > 0, \\ U(0) = U_0 & \text{a.s.} \end{cases}$$

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Definition

An approximation $(\tilde{U}_j)_j$ converges with strong order $k > 0$ if $\exists K_j > 0$:

$$E \left\{ \left| U(j\Delta t) - \tilde{U}_j \right| \right\} \leq K_j (\Delta t)^k.$$

The sample paths of the approximation $\tilde{U}$ should be close to those of the Itô process.

Numerical integration of SDE

Weak convergence

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$$\begin{cases} dU(t) = b(U, t)dt + a(U, t)dW(t), & t > 0, \\ U(0) = U_0 & \text{a.s.} \end{cases}$$

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Definition

An approximation $(\tilde{U}_j)_j$ converges with weak order $k > 0$ if for any polynomial $g \exists K_{g,j} > 0$:

$$\left| E \{g(U(j\Delta t))\} - E \{g(\tilde{U}_j)\} \right| \leq K_{g,j}(\Delta t)^k.$$

The probability distribution of the approximation should be close to that of the Itô process in order to get a good estimate of the expectation ($g(u) = u$) or the variance ($g(u) = u^2$), for example.

Time discrete approximations

Explicit 0.5-order methods

- Assume that v and a are independent of time t (thus $U(t)$ is a diffusion process), and let $t_j = j\Delta t$, $b_j = b(\tilde{U}_j)$, $a_j = a(\tilde{U}_j)$, $U_0 \sim \pi_0(du_0)$, $G \sim \mathcal{N}(0, 1)$.
- Itô SDE: the *Euler-Maruyama scheme* (1955),

$$\begin{aligned}\tilde{U}_{j+1} &= \tilde{U}_j + b_j \Delta t + a_j \sqrt{\Delta t} G, \\ \tilde{U}_0 &= U_0.\end{aligned}$$

- Stratonovich SDE: the *Euler-Heun scheme* (1982),

$$\begin{aligned}\tilde{U}_{j+1} &= \tilde{U}_j + b_j \Delta t + \tilde{a}_j \sqrt{\Delta t} G, \\ \tilde{a}_j &= \frac{1}{2} \left[a_j + a \left(\tilde{U}_j + a_j \sqrt{\Delta t} G \right) \right], \\ \tilde{U}_0 &= U_0.\end{aligned}$$

- Both have a strong order $k = \frac{1}{2}$ (vs. $k = 1$ for ordinary differential equations) and a weak order $k = 1$.

Time discrete approximations

Explicit 1-order methods

- The *Milstein scheme* (1974):

$$\begin{aligned}\tilde{U}_{j+1} &= \tilde{U}_j + b_{\lambda,j}\Delta t + a_j\sqrt{\Delta t}G + \frac{1}{2}a_ja'_j\Delta t(G^2 + 2\lambda - 1), \\ \tilde{U}_0 &= U_0,\end{aligned}$$

where $\lambda = 0$ (Itô SDE) or $\lambda = \frac{1}{2}$ (Stratonovich SDE).

- The *Runge-Kutta Milstein scheme* (1984):

$$\begin{aligned}\tilde{U}_{j+1} &= \tilde{U}_j + b_{\lambda,j}\Delta t + a_j\sqrt{\Delta t}G + \frac{1}{2}a_j\tilde{a}'_j\Delta t(G^2 + 2\lambda - 1), \\ a_j\tilde{a}'_j &= (\Delta t)^{-\frac{1}{2}} \left[a \left(\tilde{U}_j + a_j\sqrt{\Delta t} \right) - a_j \right], \\ \tilde{U}_0 &= U_0.\end{aligned}$$

- Both have strong and weak orders $k = 1$ (under mild conditions on b and a).

Time discrete approximations

Stochastic Taylor approximations

- Higher-order schemes may be derived using stochastic Taylor expansions:

$$\begin{aligned} U_{j+1} - U_j &= \int_{t_j}^{t_{j+1}} b(U) dt + \int_{t_j}^{t_{j+1}} a(U) dW \\ &\simeq \int_{t_j}^{t_{j+1}} (b(U_j) + b'(U_j) \Delta U_j) dt + \int_{t_j}^{t_{j+1}} (a(U_j) + a'(U_j) \Delta U_j) dW, \end{aligned}$$

where $\Delta U_j = \int_{t_j}^t b(U) d\tau + \int_{t_j}^t a(U) dW$.

- Then $\int_{t_j}^{t_{j+1}} \int_{t_j}^t d_\lambda W(s) d_\lambda W(t) = \frac{1}{2}(\Delta W)^2 + (\lambda - \frac{1}{2})\Delta t$.
- Higher-order expansions involve additional r.v.
 $\Delta Z_j = \int_{t_j}^{t_{j+1}} \int_{t_j}^t dW dt$ with $E\{(\Delta Z_j)^2\} \propto \Delta t^3$ etc.
- Weak Taylor approximations $U_0 \sim \hat{U}_0$, $\Delta W \sim \Delta \hat{W}$,
 $\Delta Z_j \sim \Delta \hat{Z}_j$ with approximately the same moment properties.

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Canonical equations

- $\mathbf{Q} \in \mathbb{R}^q$ the position, $\mathbf{P} \in \mathbb{R}^q$ the momentum, $\mathcal{H}$ the Hamiltonian (independent of time), $\mathbf{F}$ the non conservative forces,

$$\dot{\mathbf{Q}} = \nabla_{\mathbf{p}} \mathcal{H}(\mathbf{Q}, \mathbf{P}),$$

$$\dot{\mathbf{P}} = -\nabla_{\mathbf{q}} \mathcal{H}(\mathbf{Q}, \mathbf{P}) + \mathbf{F}(\mathbf{Q}, \mathbf{P}, \dot{\mathbf{W}}),$$

where $\mathbf{F}(\mathbf{q}, \mathbf{p}, \mathbf{f}) = -f(\mathcal{H})\mathbf{G}\nabla_{\mathbf{p}}\mathcal{H} + g(\mathcal{H})\mathbf{S}\mathbf{f}$ and $\dot{\mathbf{W}}$ a white noise.

- Example: Duffing oscillator driven by white noise,

$$M\ddot{Q} + D\dot{Q} + KQ + K_0Q^3 = g_0S_0\dot{W}$$

then $\mathcal{E}_c = \frac{1}{2}M\dot{Q}^2$, $\mathcal{E}_p = \frac{1}{2}KQ^2 + \frac{1}{4}K_0Q^4$ and $P = \partial_{\dot{Q}}\mathcal{E}_c$, thus:

$$\mathcal{H}(Q, P) = \frac{1}{2}M^{-1}P^2 + \frac{1}{2}KQ^2 + \frac{1}{4}K_0Q^4.$$

Fokker-Planck equation

- The associated *Fokker-Planck equation* for the transition PDF $\pi_t(\mathbf{q}', \mathbf{p}'; t' | \mathbf{q}, \mathbf{p}; t)$ reads:

$$\partial_t \pi + \{\pi, \mathcal{H}\} - \nabla_{\mathbf{p}} \cdot \mathbf{J}(\pi) = 0,$$

with the Poisson bracket and probability flux being defined as:

$$\begin{aligned} \{\pi, \mathcal{H}\} &= \nabla_{\mathbf{q}} \pi \cdot \nabla_{\mathbf{p}} \mathcal{H} - \nabla_{\mathbf{p}} \pi \cdot \nabla_{\mathbf{q}} \mathcal{H}, \\ \mathbf{J}(\pi) &= \pi \left[f(\mathcal{H}) \mathbf{G} + \frac{1}{2} g(\mathcal{H}) g'(\mathcal{H}) \mathbf{S} \mathbf{S}^\top \right] \nabla_{\mathbf{p}} \mathcal{H} \\ &\quad + \frac{1}{2} g(\mathcal{H})^2 \mathbf{S} \mathbf{S}^\top \nabla_{\mathbf{p}} \pi. \end{aligned}$$

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- What's new?
 - Non linear filtering of white noise,
 - A (high-dimensional) PDE for the marginal and transition PDF of diffusion processes,
 - Stochastic integrals,
 - Numerical simulations of SDE,
 - Application to non linear dynamical systems.
- What's left?
 - Numerical solutions of the FKE,
 - Computation of second-order quantities of diffusion processes.

Further reading...

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